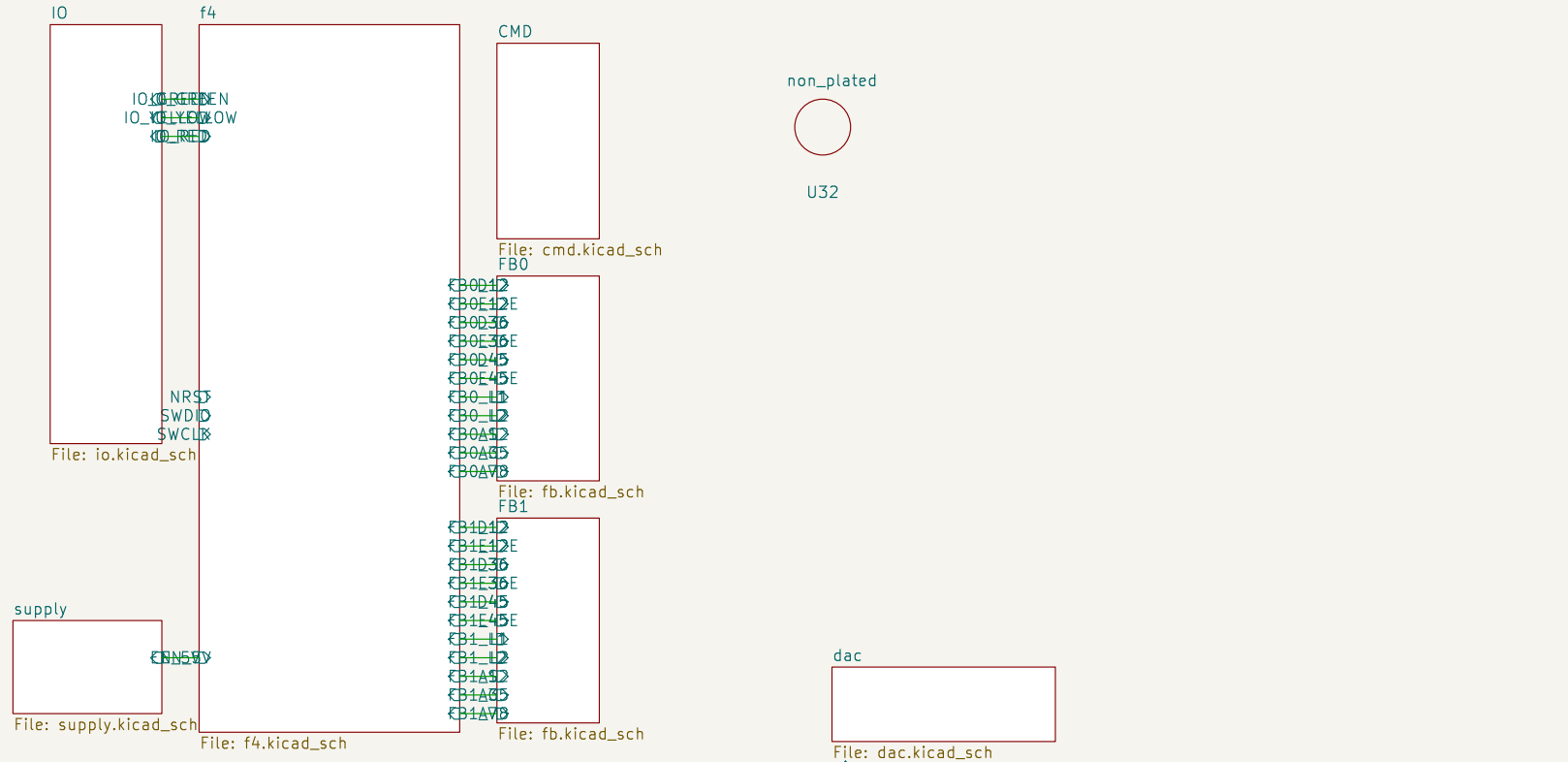
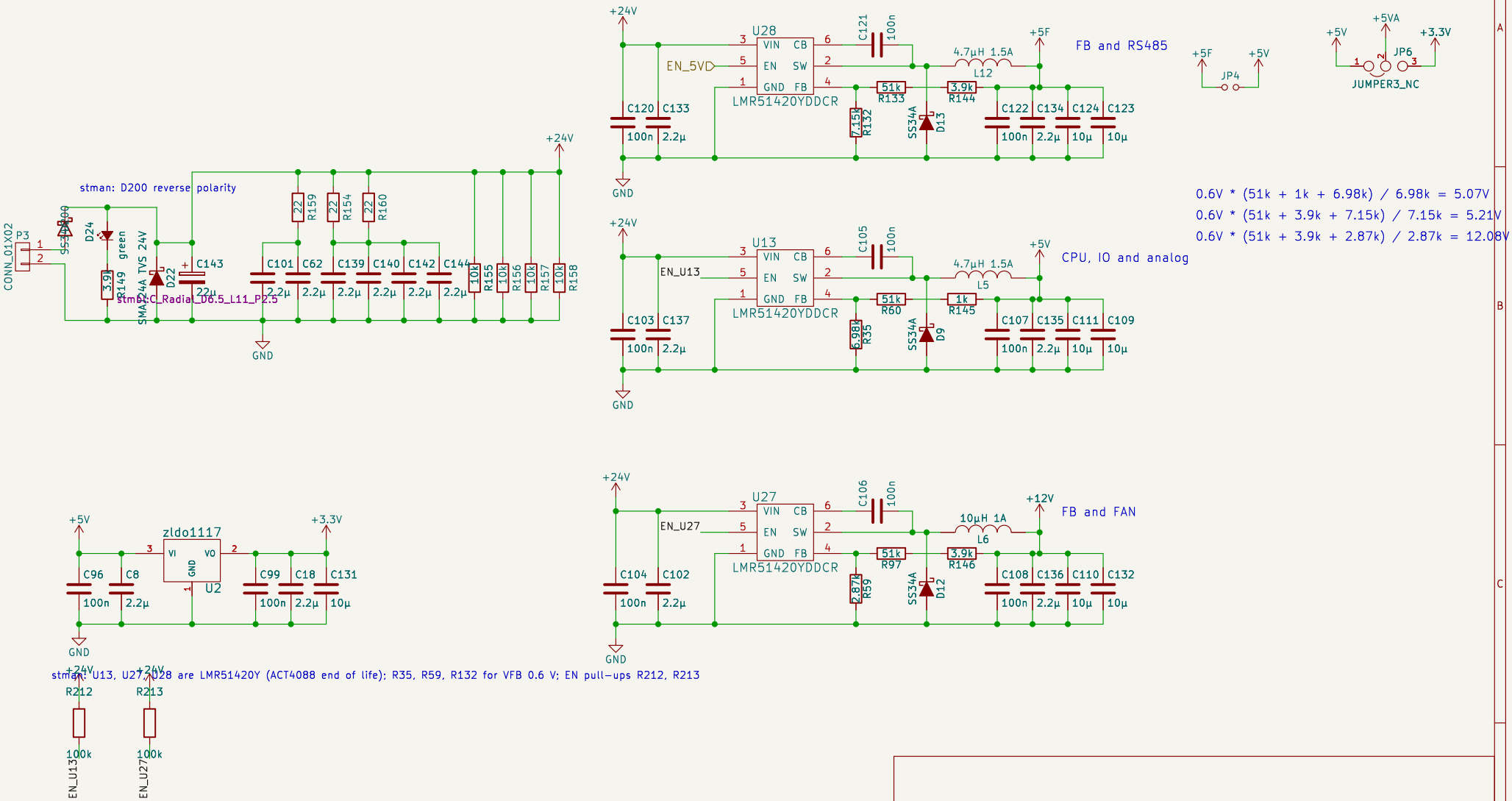


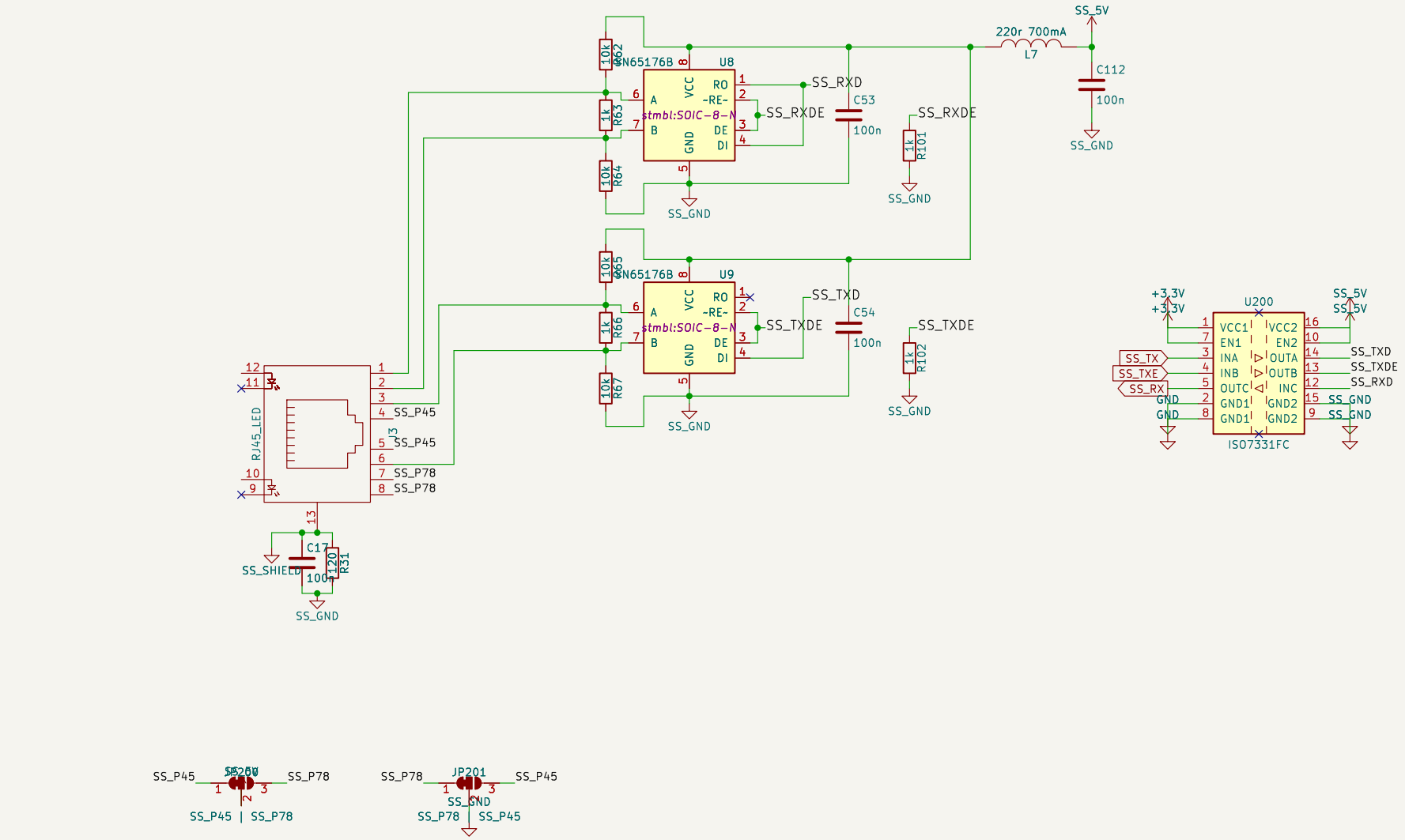
stman (docs/stman.md): stmbl v4.0's LV sheets, edited in place;
dac, ain, drvio: stman's new circuits (global labels to the MCU sheet).

Sheet: /
File: stman.kicad_sch

Title: stman: stmbl with an analog output

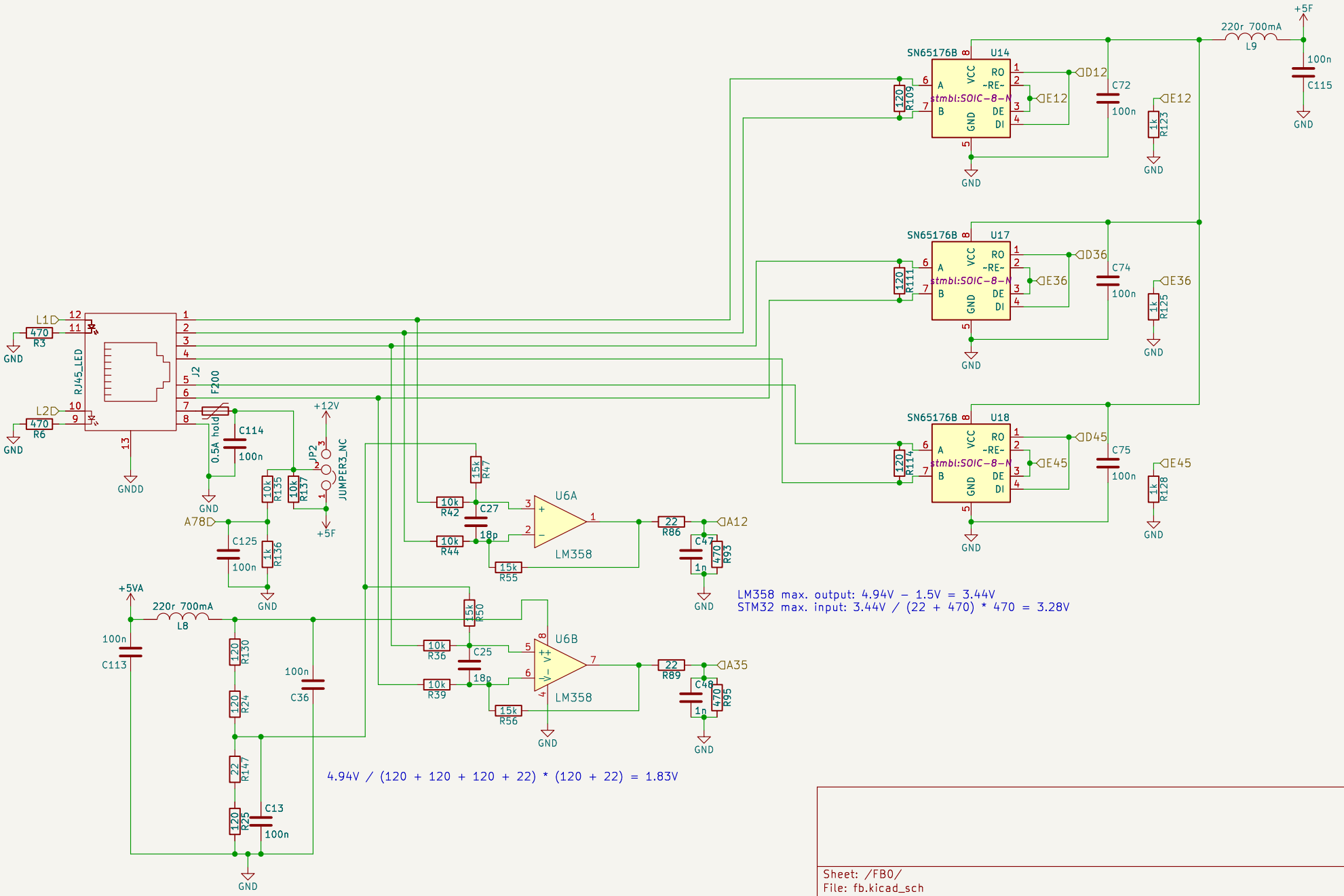
Size: A4	Date:	Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/10


stman: D200 reverse polarity
CONN_01X02
P3
SS34A
D24
green
3.9
R149
SMAJ24A TVS 24V
D22
C143
22µ
stman:CP_Radial_D6.5_L11_P2.5
22
R159
22
R154
22
R160
C101 C62 C139 C140 C142 C144
2.2µ
10k
R155
10k
R156
10k
R157
10k
R158
+24V
GND
U28
LMR51420YDDCR
VIN CB
EN SW
GND FB
EN_5VD
C121
100n
4.7µH 1.5A
L12
+5F
FB and RS485
51k
R133
3.9k
R144
7.15k
R132
SS34A
D13
C120 C133
C122 C134 C124 C123
100n 2.2µ 10µ
U13
EN_U13
C105
L5
CPU, IO and analog
51k
R60
1k
R145
6.98k
R35
D9
C103 C137
C107 C135 C111 C109
+5V
+3.3V
zldo1117
VI VO GND
U2
C96 C8
C99 C18 C131
U27
EN_U27
C106
10µH 1A
L6
+12V
FB and FAN
51k
R97
3.9k
R146
2.87k
R59
D12
C104 C102
C108 C136 C110 C132
R212
R213
100k
EN_U13
EN_U27
JP4
JP6
JUMPER3_NC
+5VA
0.6V * (51k + 1k + 6.98k) / 6.98k = 5.07V
0.6V * (51k + 3.9k + 7.15k) / 7.15k = 5.21V (stman
0.6V * (51k + 3.9k + 2.87k) / 2.87k = 12.08V
stman: U13, U27, U28 are LMR51420Y (ACT4088 end of life); R35, R59, R132 for VFB 0.6 V; EN pull-ups R212, R213
Sheet: /supply/
File: supply.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 2/10

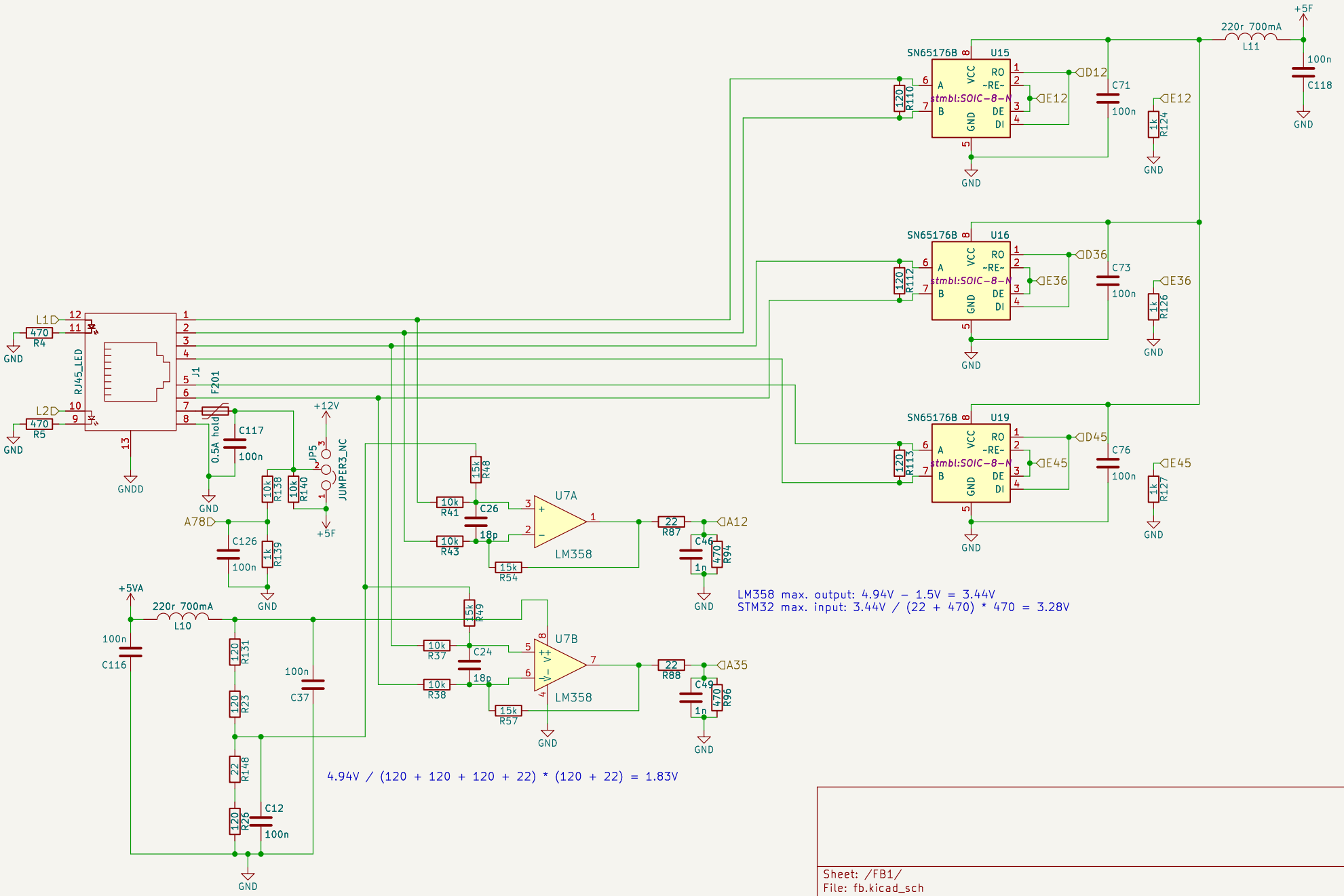


stman: the transceivers U8 (receive, pair 1/2) and U9 (send, pair 3/6) are on the cable's side,
supplied from the cable (SS_5V, SS_GND); ISO7331FC U200 (outputs low while unpowered) is the only path to the MCU.
U10, U11 (pairs 4/5, 7/8) removed: those pairs carry the cable's +5 V and GND (JP200, JP201: check against the Mesa card *)

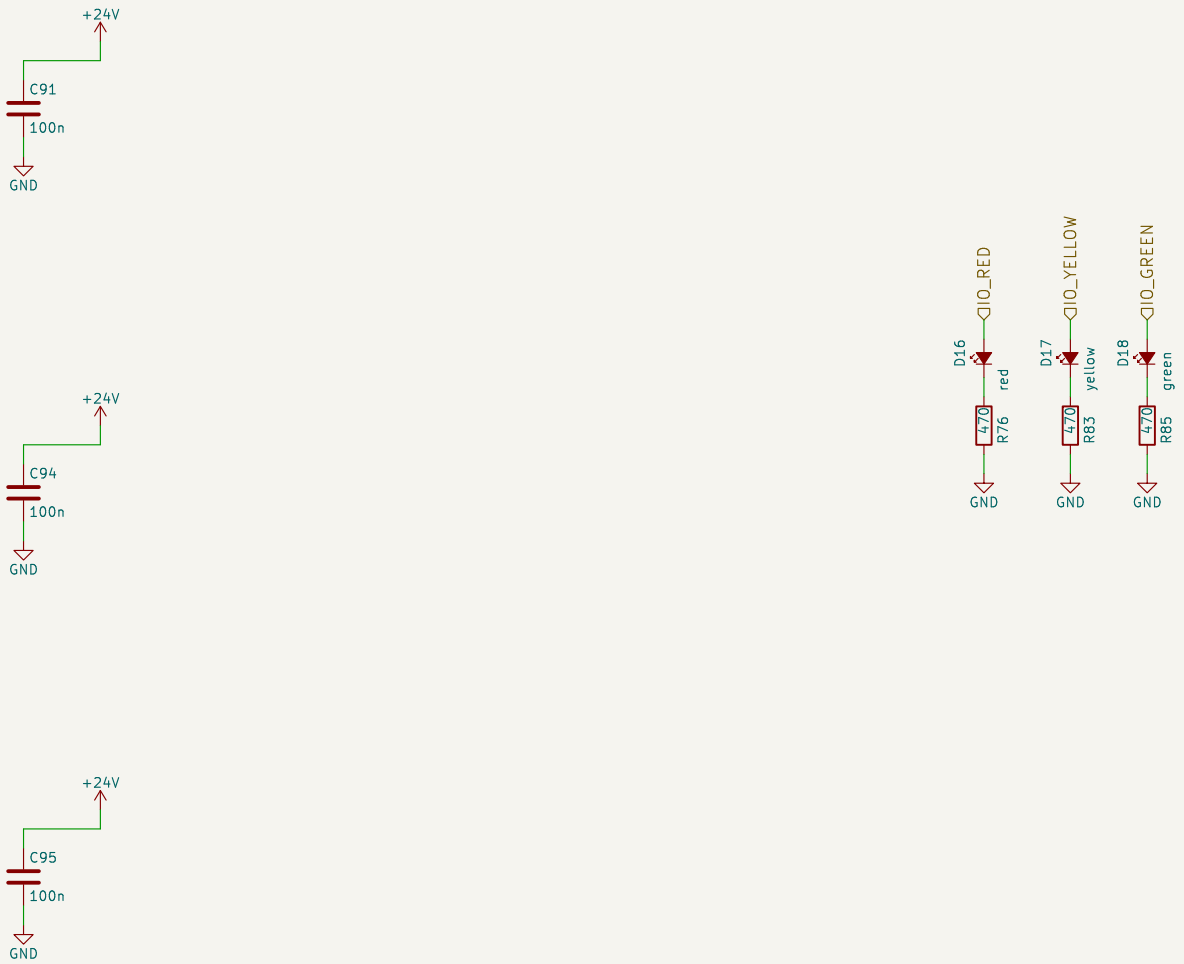
Sheet: /CMD/ File: cmd.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/10



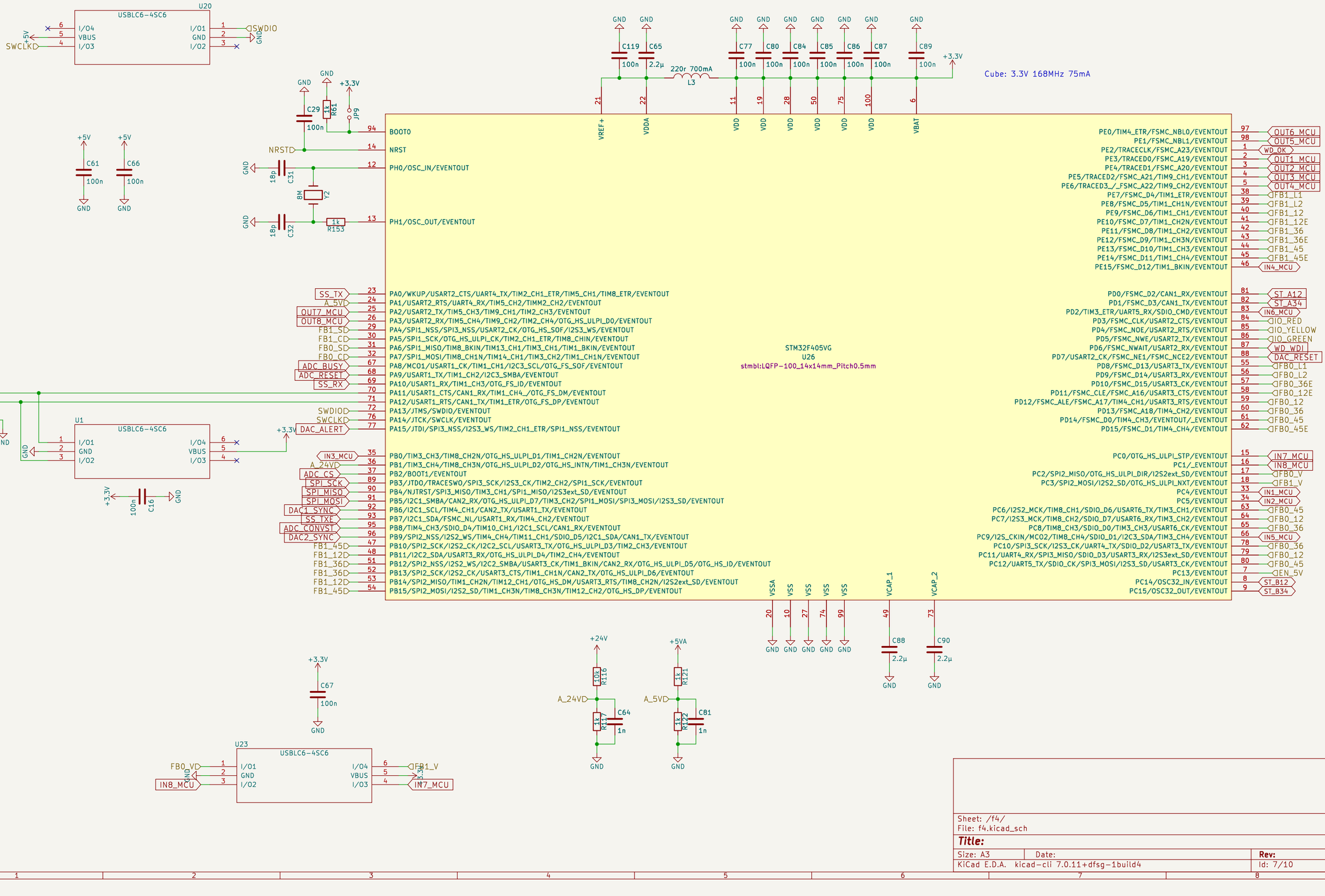
Sheet: /FB0/		
File: fb.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/10



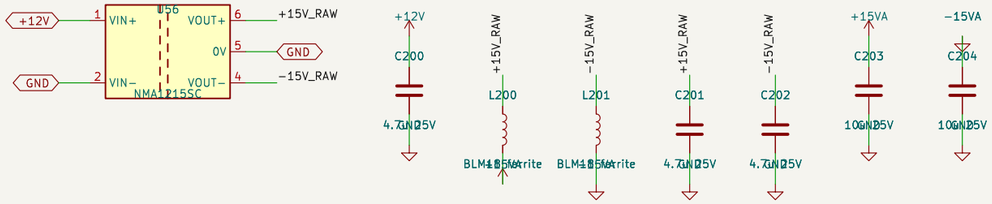
Sheet: /FB1/		
File: fb.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/10



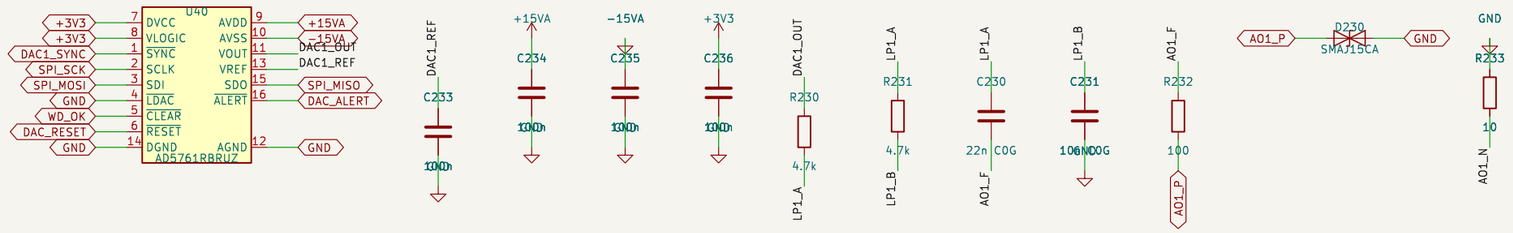
Sheet: /IO/		
File: io.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/10


U20
USBLC6-4SC6
I/O4
VBUS
I/O3
I/O1
GND
I/O2
SWCLK
SWDIO
+5V
GND
C61
C66
100n
C29
R61
1k
JP9
NRST
C31
C32
18p
8M
Y2
R153
BOOT0
NRST
PH0/OSC_IN/EVENTOUT
PH1/OSC_OUT/EVENTOUT
C119
C65
2.2µ
220r 700mA
L3
C77
C80
C84
C85
C86
C87
C89
+3.3V
Cube: 3.3V 168MHz 75mA
VREF+
VDDA
VDD
VBAT
STM32F405VG
U26
stmbl:LQFP-100_14x14mm_Pitch0.5mm
PA0/WKUP/USART2_CTS/UART4_TX/TIM2_CH1_ETR/TIM5_CH1/TIM8_ETR/EVENTOUT
PA1/USART2_RTS/UART4_RX/TIM5_CH2/TIMM2_CH2/EVENTOUT
PA2/USART2_TX/TIM5_CH3/TIM9_CH1/TIM2_CH3/EVENTOUT
PA3/USART2_RX/TIM5_CH4/TIM9_CH2/TIM2_CH4/OTG_HS_ULPI_D0/EVENTOUT
PA4/SPI1_NSS/SPI3_NSS/USART2_CK/OTG_HS_SOF/I2S3_WS/EVENTOUT
PA5/SPI1_SCK/OTG_HS_ULPI_CK/TIM2_CH1_ETR/TIM8_CHIN/EVENTOUT
PA6/SPI1_MISO/TIM8_BKIN/TIM13_CH1/TIM3_CH1/TIM1_BKIN/EVENTOUT
PA7/SPI1_MOSI/TIM8_CH1N/TIM14_CH1/TIM3_CH2/TIM1_CH1N/EVENTOUT
PA8/MCO1/USART1_CK/TIM1_CH1/I2C3_SCL/OTG_FS_SOF/EVENTOUT
PA9/USART1_TX/TIM1_CH2/I2C3_SMBA/EVENTOUT
PA10/USART1_RX/TIM1_CH3/OTG_FS_ID/EVENTOUT
PA11/USART1_CTS/CAN1_RX/TIM1_CH4_/OTG_FS_DM/EVENTOUT
PA12/USART1_RTS/CAN1_TX/TIM1_ETR/OTG_FS_DP/EVENTOUT
PA13/JTMS/SWDIO/EVENTOUT
PA14/JTCK/SWCLK/EVENTOUT
PA15/JTDI/SPI3_NSS/I2S3_WS/TIM2_CH1_ETR/SPI1_NSS/EVENTOUT
PB0/TIM3_CH3/TIM8_CH2N/OTG_HS_ULPI_D1/TIM1_CH2N/EVENTOUT
PB1/TIM3_CH4/TIM8_CH3N/OTG_HS_ULPI_D2/OTG_HS_INTN/TIM1_CH3N/EVENTOUT
PB2/BOOT1/EVENTOUT
PB3/JTDO/TRACESWO/SPI3_SCK/I2S3_CK/TIM2_CH2/SPI1_SCK/EVENTOUT
PB4/NJTRST/SPI3_MISO/TIM3_CH1/SPI1_MISO/I2S3ext_SD/EVENTOUT
PB5/I2C1_SMBA/CAN2_RX/OTG_HS_ULPI_D7/TIM3_CH2/SPI1_MOSI/SPI3_MOSI/I2S3_SD/EVENTOUT
PB6/I2C1_SCL/TIM4_CH1/CAN2_TX/USART1_TX/EVENTOUT
PB7/I2C1_SDA/FSMC_NL/USART1_RX/TIM4_CH2/EVENTOUT
PB8/TIM4_CH3/SDIO_D4/TIM10_CH1/I2C1_SCL/CAN1_RX/EVENTOUT
PB9/SPI2_NSS/I2S2_WS/TIM4_CH4/TIM11_CH1/SDIO_D5/I2C1_SDA/CAN1_TX/EVENTOUT
PB10/SPI2_SCK/I2S2_CK/I2C2_SCL/USART3_TX/OTG_HS_ULPI_D3/TIM2_CH3/EVENTOUT
PB11/I2C2_SDA/USART3_RX/OTG_HS_ULPI_D4/TIM2_CH4/EVENTOUT
PB12/SPI2_NSS/I2S2_WS/I2C2_SMBA/USART3_CK/TIM1_BKIN/CAN2_RX/OTG_HS_ULPI_D5/OTG_HS_ID/EVENTOUT
PB13/SPI2_SCK/I2S2_CK/USART3_CTS/TIM1_CH1N/CAN2_TX/OTG_HS_ULPI_D6/EVENTOUT
PB14/SPI2_MISO/TIM1_CH2N/TIM12_CH1/OTG_HS_DM/USART3_RTS/TIM8_CH2N/I2S2ext_SD/EVENTOUT
PB15/SPI2_MOSI/I2S2_SD/TIM1_CH3N/TIM8_CH3N/TIM12_CH2/OTG_HS_DP/EVENTOUT
PE0/TIM4_ETR/FSMC_NBL0/EVENTOUT
PE1/FSMC_NBL1/EVENTOUT
PE2/TRACECLK/FSMC_A23/EVENTOUT
PE3/TRACED0/FSMC_A19/EVENTOUT
PE4/TRACED1/FSMC_A20/EVENTOUT
PE5/TRACED2/FSMC_A21/TIM9_CH1/EVENTOUT
PE6/TRACED3_/_FSMC_A22/TIM9_CH2/EVENTOUT
PE7/FSMC_D4/TIM1_ETR/EVENTOUT
PE8/FSMC_D5/TIM1_CH1N/EVENTOUT
PE9/FSMC_D6/TIM1_CH1/EVENTOUT
PE10/FSMC_D7/TIM1_CH2N/EVENTOUT
PE11/FSMC_D8/TIM1_CH2/EVENTOUT
PE12/FSMC_D9/TIM1_CH3N/EVENTOUT
PE13/FSMC_D10/TIM1_CH3/EVENTOUT
PE14/FSMC_D11/TIM1_CH4/EVENTOUT
PE15/FSMC_D12/TIM1_BKIN/EVENTOUT
PD0/FSMC_D2/CAN1_RX/EVENTOUT
PD1/FSMC_D3/CAN1_TX/EVENTOUT
PD2/TIM3_ETR/UART5_RX/SDIO_CMD/EVENTOUT
PD3/FSMC_CLK/USART2_CTS/EVENTOUT
PD4/FSMC_NOE/USART2_RTS/EVENTOUT
PD5/FSMC_NWE/USART2_TX/EVENTOUT
PD6/FSMC_NWAIT/USART2_RX/EVENTOUT
PD7/USART2_CK/FSMC_NE1/FSMC_NCE2/EVENTOUT
PD8/FSMC_D13/USART3_TX/EVENTOUT
PD9/FSMC_D14/USART3_RX/EVENTOUT
PD10/FSMC_D15/USART3_CK/EVENTOUT
PD11/FSMC_CLE/FSMC_A16/USART3_CTS/EVENTOUT
PD12/FSMC_ALE/FSMC_A17/TIM4_CH1/USART3_RTS/EVENTOUT
PD13/FSMC_A18/TIM4_CH2/EVENTOUT
PD14/FSMC_D0/TIM4_CH3/EVENTOUT/_EVENTOUT
PD15/FSMC_D1/TIM4_CH4/EVENTOUT
PC0/OTG_HS_ULPI_STP/EVENTOUT
PC1/_EVENTOUT
PC2/SPI2_MISO/OTG_HS_ULPI_DIR/I2S2ext_SD/EVENTOUT
PC3/SPI2_MOSI/I2S2_SD/OTG_HS_ULPI_NXT/EVENTOUT
PC4/EVENTOUT
PC5/EVENTOUT
PC6/I2S2_MCK/TIM8_CH1/SDIO_D6/USART6_TX/TIM3_CH1/EVENTOUT
PC7/I2S3_MCK/TIM8_CH2/SDIO_D7/USART6_RX/TIM3_CH2/EVENTOUT
PC8/TIM8_CH3/SDIO_D0/TIM3_CH3/USART6_CK/EVENTOUT
PC9/I2S_CKIN/MCO2/TIM8_CH4/SDIO_D1/I2C3_SDA/TIM3_CH4/EVENTOUT
PC10/SPI3_SCK/I2S3_CK/UART4_TX/SDIO_D2/USART3_TX/EVENTOUT
PC11/UART4_RX/SPI3_MISO/SDIO_D3/USART3_RX/I2S3ext_SD/EVENTOUT
PC12/UART5_TX/SDIO_CK/SPI3_MOSI/I2S3_SD/USART3_CK/EVENTOUT
PC13/EVENTOUT
PC14/OSC32_IN/EVENTOUT
PC15/OSC32_OUT/EVENTOUT
VSSA
VSS
VCAP_1
VCAP_2
C88
C90
2.2µ
SS_TX
A_5V
OUT7_MCU
OUT8_MCU
FB1_SD
FB1_CD
FB0_SD
FB0_CD
ADC_BUSY
ADC_RESET
SS_RX
SWDIO
SWCLK
DAC_ALERT
IN3_MCU
A_24V
ADC_CS
SPI_SCK
SPI_MISO
SPI_MOSI
DAC1_SYNC
SS_TXE
ADC_CONVST
DAC2_SYNC
FB1_45
FB1_12
FB1_36
FB1_36
FB1_12
FB1_45
OUT6_MCU
OUT5_MCU
WD_OK
OUT1_MCU
OUT2_MCU
OUT3_MCU
OUT4_MCU
FB1_L1
FB1_L2
FB1_12
FB1_12E
FB1_36
FB1_36E
FB1_45
FB1_45E
IN4_MCU
ST_A12
ST_A34
IN6_MCU
IO_RED
IO_YELLOW
IO_GREEN
WD_WDI
DAC_RESET
FB0_L1
FB0_L2
FB0_36E
FB0_12E
FB0_12
FB0_36
FB0_45
FB0_45E
IN7_MCU
IN8_MCU
FB0_V
FB1_V
IN1_MCU
IN2_MCU
FB0_45
FB0_12
FB0_36
IN5_MCU
FB0_36
FB0_12
FB0_45
EN_5V
ST_B12
ST_B34
USB_OTG
VCC
D-
D+
ID
GND
shield
P1
C1
100n
120
R1
U1
USBLC6-4SC6
C16
U23
FB0_V
IN8_MCU
FB1_V
IN7_MCU
C67
+24V
+5VA
10Ω
R116
1k
R117
C64
1n
R121
R122
C81
Sheet: /f4/
File: f4.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 7/10

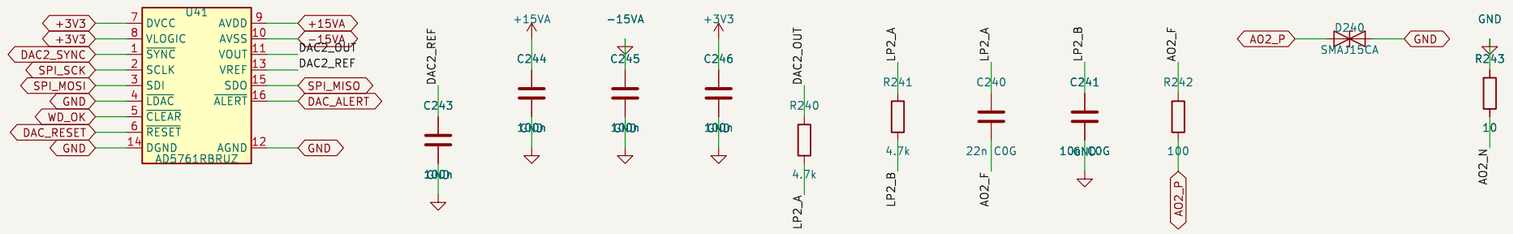
±15 V: NMA1215 from +12 V, LC filtered



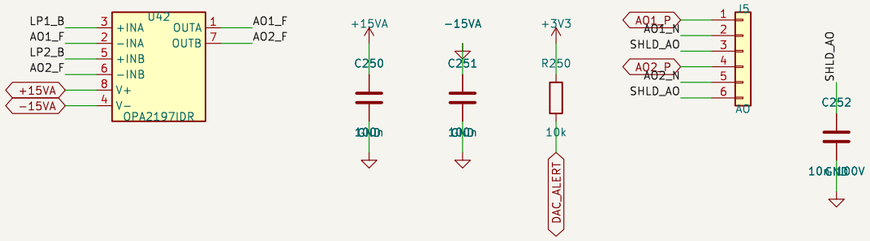
AO1: DAC, Sallen–Key (R230 R231 C230 C231, U42A), 100 R, TVS



AO2 (fitted for the spindle's C axis): DAC, Sallen–Key (U42B)



Filter op–amp, /ALERT, connector J5



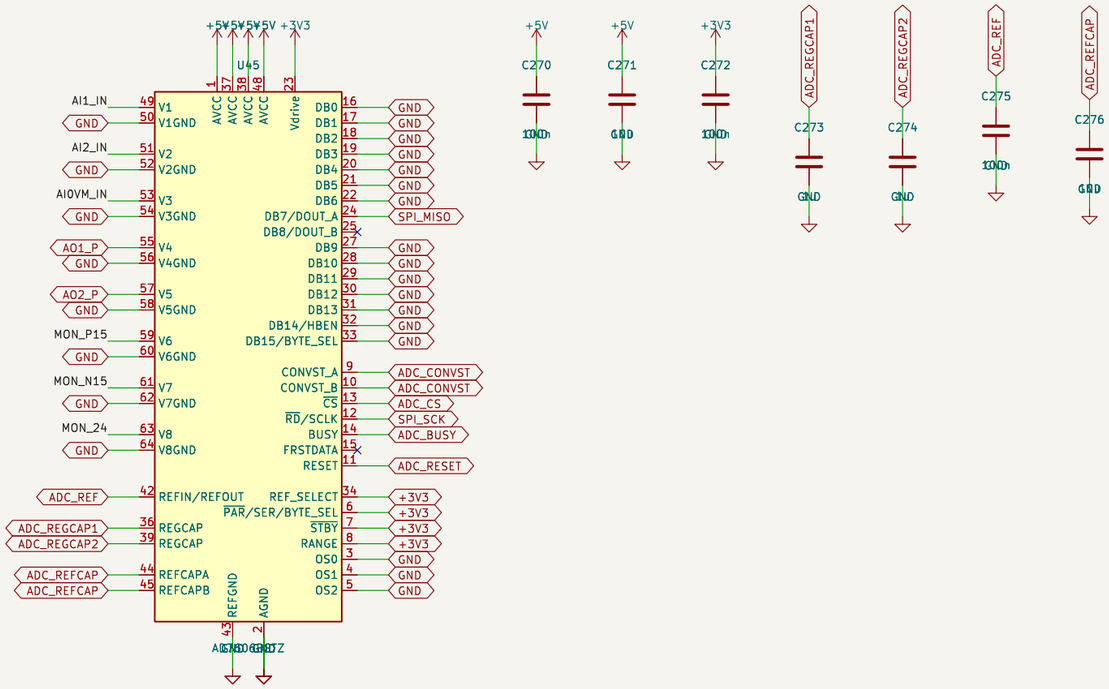
* in a part's Note: check against the datasheet
docs/stman.md

Sheet: /dac/
File: dac.kicad_sch

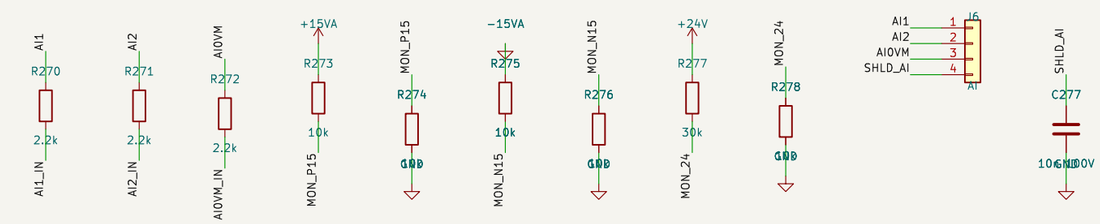
Title: stman: Command outputs AO1, AO2: AD5761R, 2nd–order low–pass 2.3

Size: A3	Date:	**Rev:**
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4		Id: 10/10

AD7606: 8 ch simultaneous, ±10 V (RANGE high), serial (PAR/SER high), internal reference



Inputs: 2.2k series (the ADC's 1 MΩ inputs, ±16.5 V clamps); rail dividers; J6



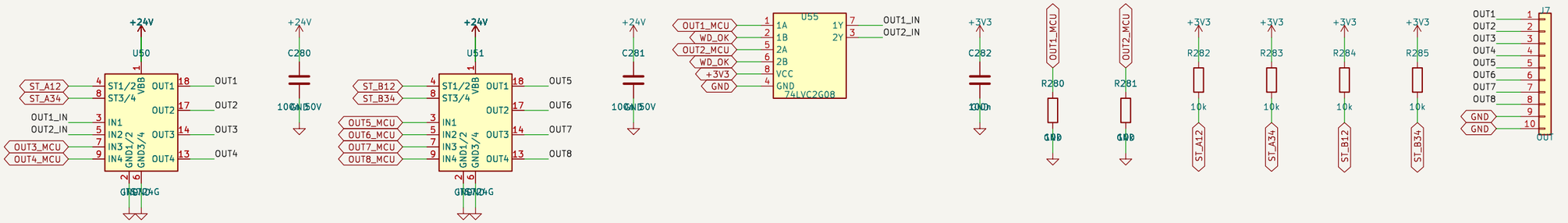
* in a part's Note: check against the datasheet
docs/stman.md

Sheet: /ain/
File: ain.kicad_sch

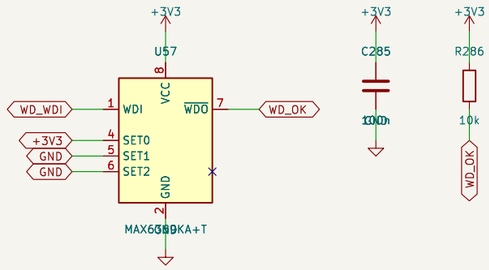
Title: stman: Monitor ADC: AD7606 (drive monitors and 0VM, AO readbacks, r

Size: A3	Date:	**Rev:**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 11/10

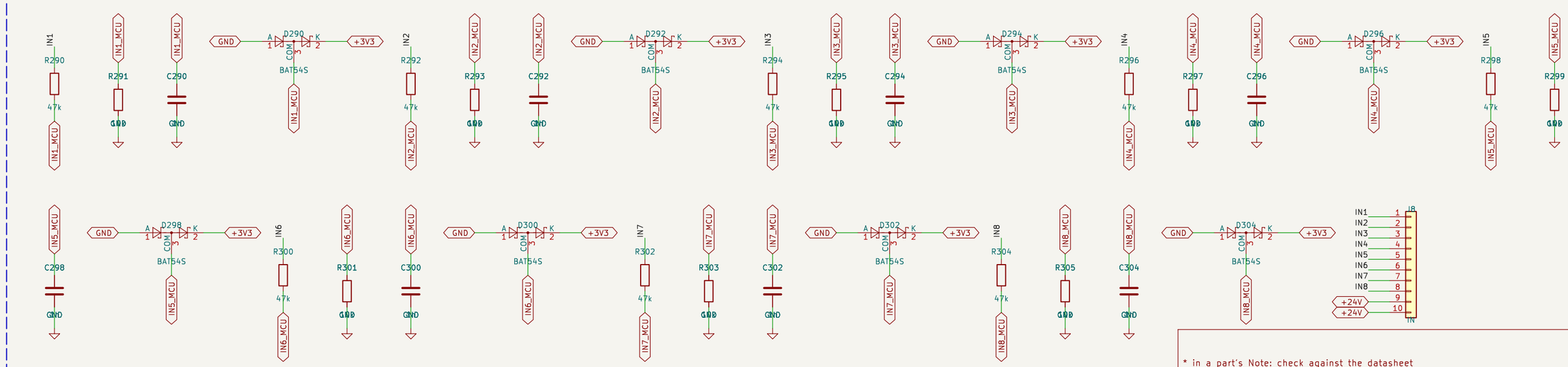
Outputs: ITS724G x2, OUT1/OUT2 gated by the watchdog



Watchdog MAX6369: WD_OK low -> DACs to 0 V, OUT1/OUT2 off



Inputs: 47k / 10k, 1 nF, BAT54S clamp (high > ~13 V, low < ~6 V)



* in a part's Note: check against the datasheet docs/stman.md

Sheet: /drvio/
File: drvio.kicad_sch

Title: stman: Drive I/O: 8 x 24 V out (ITS724G), 8 x 24 V in, watchdog

Size: A3	Date:	Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 12/10